

VHDL Assignment 1

NOT Gate in VHDL, Simulated with Logisim-evolution and Questa

Name: _____ Student ID: _____

1. Objective

The goal of this assignment is to become familiar with the VHDL workflow: writing a simple VHDL entity, importing it into Logisim-evolution, and simulating it with Questa. A NOT gate was implemented and verified.

2. VHDL Code (not_gate.vhd)

```
library ieee;
use ieee.std_logic_1164.all;

entity not_gate is
    port (
        a : in  std_logic;
        y : out std_logic
    );
end not_gate;

architecture behavior of not_gate is
begin
    y <= not a;
end behavior;
```

The entity **not_gate** has a 1-bit input **a** and a 1-bit output **y**. The architecture uses a single concurrent assignment, `y <= not a;`, so the output is always the logical inverse of the input.

3. Circuit in Logisim-evolution

The VHDL file was imported with the VHDL import function, and the **not_gate** component was placed in the **main** circuit. An input pin labeled **a** was connected to port a, and an output pin labeled **y** was connected to port y. The component label was set to U1, as required for HDL components. The Questa path was set in File > Preferences > Software.

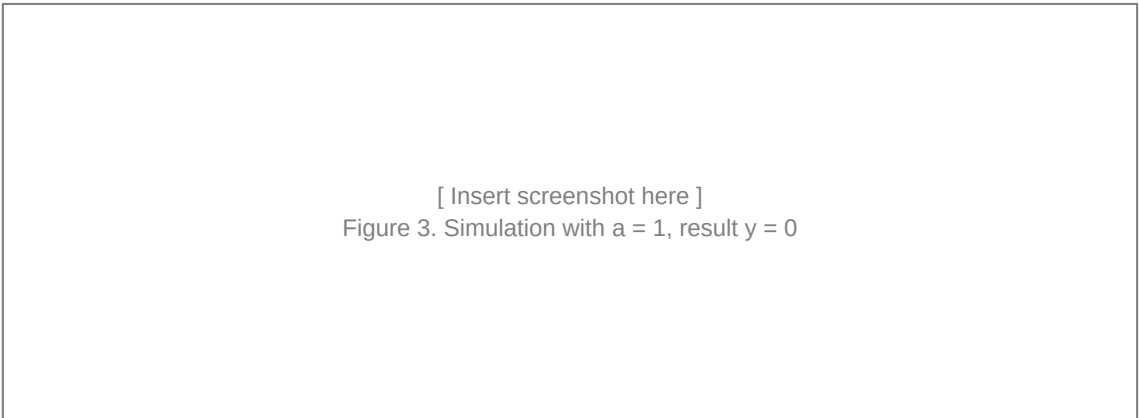
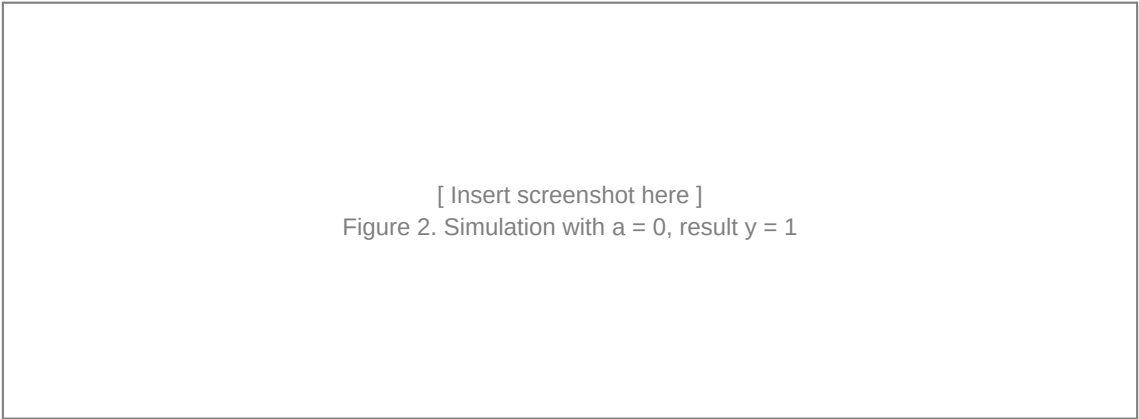
[Insert screenshot here]

Figure 1. Circuit: input pin a, not_gate block, output pin y

4. Simulation Results

The input was toggled with the Poke tool, and the simulation was advanced with Tick (Ctrl+T) until the output updated.

a (input)	y (output)
0	1
1	0



5. Explanation of Results

A NOT gate outputs the opposite of its input. When **a = 0**, the simulation shows **y = 1** (Figure 2). When **a = 1**, the simulation shows **y = 0** (Figure 3). Both results match the truth table of a NOT gate, which confirms that the VHDL description is correct and that the Logisim-evolution and Questa workflow works as expected.

6. Submitted Files

not_gate.vhd (VHDL source), not_gate.circ (Logisim project), and this PDF report.